

Bounded operators on Hilbert spaces.

• Examples of weak convergence. $\{e_\alpha\}_{\alpha \in I}$ basis of $\mathcal{H}$. Then:

$$x_n \rightarrow x \iff \begin{cases} \|x_n\| \leq M & ; \text{ and } \forall \alpha \in I, \\ \langle x_n, e_\alpha \rangle \rightarrow \langle x, e_\alpha \rangle & \forall \alpha \in I. \end{cases}$$

• proof $y = \sum_{\alpha \in I \text{ finite}} \langle e_\alpha, y \rangle e_\alpha$ forms a dense set $\mathcal{D}$ in $\mathcal{H}$

clearly, $\langle x_n, y \rangle \rightarrow \langle x, y \rangle$ for such $y \in \mathcal{D}$.

• We saw $\|x_n\| \leq M$ was necessary. Indeed take:

$$n e_n = (0, \dots, 0, n, 0, \dots). \quad \langle n e_n, e_\alpha \rangle \rightarrow 0 \text{ as } n \rightarrow \infty.$$

and yet $n e_n$ does not converge weakly to 0 (check!).

$$(\text{try } y = \sum_{m \geq 1} \frac{1}{m} e_m \in \mathcal{H})$$

• Example $X_n = \sin(n\pi x)$ in $\mathcal{H} = L^2(0, 1)$

• we want to show $X_n \rightarrow 0$ i.e., $\int_0^1 \sin(n\pi x) f(x) dx \rightarrow 0 \quad \forall f \in L^2(0, 1)$.

• We can test this on basis e_α with $e_\alpha(x) = e^{im2\pi x}$

• Indeed $\int_0^1 \sin(n\pi x) e^{im2\pi x} dx = \langle X_n, e_m \rangle \rightarrow 0$ as $n \rightarrow \infty$.

• $\|X_n\| \leq 1$ so bounded $\checkmark$. Thus $X_n \rightarrow 0$ weakly.

• Again, $X_n \not\rightarrow 0$ strong since $\|X_n\|^2 = \|\sin n\pi x\|^2 \not\rightarrow 0$.

• Proposition $X_n \rightarrow x \Rightarrow \|x\| \leq \liminf_{n \rightarrow \infty} \|X_n\|$.

• $X_n \rightarrow x$ and $\|X_n\| \rightarrow \|x\| \Rightarrow X_n \rightarrow x$ strong.

• Proof $\|x\|^2 = \langle x, x \rangle = \lim_{n \rightarrow \infty} \langle x, X_n \rangle \leq \liminf_{n \rightarrow \infty} \|x\| \|X_n\|$.

$$\|x - X_n\|^2 = \|x\|^2 + \|X_n\|^2 - 2 \langle x, X_n \rangle \rightarrow 0 \text{ as } n \rightarrow \infty. \quad \square$$

$(\cdot, \cdot) \rightarrow \mathbb{R}$
 $\downarrow$
 $\|x\|^2$
 $\downarrow$
 $\|x\|^2$

Theorem Banach Alaoglu. The closed unit ball is weakly compact.

- So, $\|x_n\| \leq 1 \Rightarrow x_{\varphi(n)} \rightarrow x \in \mathcal{H}$.
- proof of weak sequential compactness in separable Hilbert space. $\{e_m\}$ o.n.b.
- $|\langle x_n, e_m \rangle| \leq 1$ so $\langle x_{\varphi(n)}, e_1 \rangle \xrightarrow{n \rightarrow \infty} c_1$ by Heine - Borel.
iteratively, $\langle x_{\varphi_2 \circ \dots \circ \varphi_1(n)}, e_k \rangle \xrightarrow{n \rightarrow \infty} c_k$. Define $\varphi(y) := \lim_{k \rightarrow \infty} \langle x_{\varphi_2 \circ \dots \circ \varphi_1(k)}, y \rangle$.
- φ is defined since it is clearly defined on dense set $y = \sum_{k=1}^K \langle e_k, y \rangle e_k$
and $|\varphi(y)| \leq \|y\|$ there. Use B.L.T or Hahn Banach to define it on $\mathcal{H}$.
- Thus, $\|\varphi\| \leq 1$. By the Riesz representation, $\varphi(y) = \langle x, y \rangle$.
Therefore a subsequence of $\langle x_n, y \rangle$ converges to $\langle x, y \rangle$
with the subsequence chosen independent of y . So $x_n \rightarrow x \quad \square$
- See [H.v.] for applications of weak convergence to minimization problems.

Spectrum of bounded linear operators (Chapter 9).

Operators on $\mathbb{C}^n$: matrix A . Look for spectral decomposition $Au = \lambda u$; $(A - \lambda I)u = 0$.

Assume $Au_k = \lambda_k u_k$ $\lambda_k \in \mathbb{C}$ with $\{u_k\}_{1 \leq k \leq n}$ an o.n.b. of $\mathbb{C}^n$.

$\sigma(A) := \{\lambda_k\}_{1 \leq k \leq n}$ is the spectrum of A (where $A - \lambda I$ not invertible).

collect $U = \begin{pmatrix} | & & | \\ u_1 & \dots & u_n \\ | & & | \end{pmatrix}$ so that $Ue_k = u_k$ and $U^*u_k = e_k$.

Thus U is unitary with $U^* = U^{-1}$. Now observe that:

$$(U^*AU)e_k = \lambda_k e_k \quad U^*AU = D \Leftrightarrow A = UDU^*.$$

Observe: $AA^* = A^*A$ $A^* = U\bar{D}U^*$, $AA^* = A^*A = U D \bar{D} U^*$.

Theorem: A is normal iff $\mathbb{C}^n$ has an orthonormal basis of eigenvectors of A .

In finite dimensions, we use $P_A(\lambda) = \det(A - \lambda I)$ with $P_A(\lambda) = 0 \Leftrightarrow \ker(A - \lambda I) \neq \{0\}$.

All polynomials have roots $\leadsto$ spectral decomposition. $\left\{ \begin{array}{l} \text{w.B. } A = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \text{ not normal;} \\ \text{cannot be diagonalized} \end{array} \right.$

Let $\mathcal{H}$ be Hilbert space and $\mathcal{B}(\mathcal{H})$ the space of bounded operators.

Definition: * the resolvent set of $A \in \mathcal{B}(\mathcal{H})$ is $\rho(A) \subset \mathbb{C}$ the set of $\lambda \in \mathbb{C}$ such that $(A - \lambda I) : \mathcal{H} \rightarrow \mathcal{H}$ is one-to-one and onto.

* the spectrum of A is $\mathbb{C} \setminus \rho(A) =: \sigma(A)$.

Open mapping implies $(A - \lambda I)^{-1}$ is bounded for $\lambda \in \rho(A)$.

λ is an eigenvalue of A if $Au = \lambda u$ for $0 \neq u \in \mathcal{H}$.

$(A - \lambda I)$ is not one-to-one. Then $\lambda \in \sigma(A)$ but $\sigma(A)$ is richer than eigenvalues.

Definition (a) point spectrum $\lambda \in \sigma(A)$ s.t. $(A - \lambda I)$ is not one-to-one. Then λ is indeed an eigenvalue.

$\sigma = \sigma_p \cup \sigma_c$ (b) Continuous spectrum if $(A - \lambda I)$ is one-to-one but not onto and its range is dense in $\mathcal{H}$: $\overline{\text{ran}(A - \lambda I)} = \mathcal{H}$.

$\cup \sigma_r$ (c) Residual spectrum if $(A - \lambda I)$ is one-to-one, not onto, and $\overline{\text{ran}(A - \lambda I)} \neq \mathcal{H}$. This means $\overline{\text{ran}(A - \lambda I)} \oplus \mathcal{H}^\perp = \mathcal{H}$ and $\mathcal{H}^\perp \neq \{0\}$.

. Examples on $\mathcal{H} = L^2(0,1)$. $Mf(x) = xf(x)$; $M \in \mathcal{B}(\mathcal{H})$.

. $(M - x_0)f(x) = (x - x_0)f(x)$ "inverse" $(M - x_0)^{-1} = (x - x_0)^{-1}$

is problematic when $x = x_0$.

. Heuristically $(M - x_0)\delta_{x_0}(x) = (x - x_0)\delta_{x_0}(x) = 0$

So x_0 looks like a generalized eigenvalue with δ_{x_0} a generalized eigenvector.

. We can show that $\sigma(M) = \sigma_c(M) = [0, 1]$.

. Indeed, for $x_0 \in [0, 1]$, $(M - x_0)$ is not onto since $(x - x_0)^{-1} \notin L^2(0,1)$.

. $1 \notin \text{ran}(M - x_0)$. For $f \in L^2(0,1)$, define $f_n(x) = f(x) \chi(|x - x_0| \geq \frac{1}{n})$.

. $f_n \rightarrow f$ in $L^2(0,1)$ and $f_n \in \text{ran}(M - x_0)$. So, $\overline{\text{ran}(M - x_0)} = L^2(0,1)$.

. For $x_0 \notin [0, 1]$, $(x - x_0)^{-1}$ clearly bounded. $(M - x_0)$ clearly one-to-one $\square$

. Definition. Resolvent operator, defined for $\lambda \in \rho(A)$ by $R_\lambda = (\lambda I - A)^{-1}$
 $= \frac{1}{\lambda} \left(I - \frac{A}{\lambda} \right)^{-1}$.

$R_\lambda \in \mathcal{B}(\mathcal{H})$.

Resolvent $R_\lambda = (\lambda I - A)^{-1} = \frac{1}{\lambda} (I - \frac{1}{\lambda} A)^{-1}$. $\lambda \in \rho(A)$.

Proposition: $\mathbb{C} \setminus \{ \lambda \in \mathbb{C}, |\lambda| \leq \|A\| \} = \{ \lambda \in \mathbb{C}, |\lambda| > \|A\| \} \subset \rho(A)$.

R_λ is analytic in λ on $\rho(A)$. $\rho(A)$ is open in $\mathbb{C}$.

proof: $\lambda_0 \in \rho(A)$ $\lambda = \lambda - \lambda_0 + \lambda_0$. $\lambda I - A = (\lambda_0 - A) \left(I - \underbrace{(\lambda_0 - \lambda)(\lambda_0 - A)^{-1}}_{K_\lambda} \right) = (\lambda_0 - A)(I - K_\lambda)$.

For $|\lambda - \lambda_0| \|(\lambda_0 - A)^{-1}\| < 1$, we have $\|K_\lambda\| < 1 \Rightarrow (\lambda - A)^{-1} = (I - K_\lambda)^{-1} (\lambda_0 - A)^{-1}$

This implies $\rho(A)$ is open.

Moreover, $R_\lambda = \sum_{k=0}^{\infty} (K_\lambda)^k (\lambda_0 - A)^{-1}$ Convergent expansion in λ
 $\Rightarrow R_\lambda$ is analytic.

Finally, $R_\lambda = \frac{1}{\lambda} (I - \frac{A}{\lambda})^{-1}$. For $|\lambda| > \|A\|$, $\|\frac{A}{\lambda}\| < 1$

and $R_\lambda = \frac{1}{\lambda} \sum_{k=0}^{\infty} \left(\frac{A}{\lambda}\right)^k$ bounded $\Rightarrow \lambda \in \rho(A)$.

Thus $\sigma(A) \subset \{ |\lambda| \leq \|A\| \}$ and $\sigma(A)$ is closed.

• Definition spectral radius $r(A) = \sup \{ |\lambda|, \lambda \in \sigma(A) \}$.

• clearly, $r(A) \leq \|A\|$. Proposition (see [H.W.]) $r(A) = \lim_{n \rightarrow \infty} \|A^n\|^{\frac{1}{n}}$.

• the proofs not immediate; see [H.W. and bibliography].

• when $A = A^*$, then $r(A) = \|A\|$. This is easier: $\|A^2\| = \|A\|^2 \Rightarrow \|A^{2^n}\| = \|A\|^{2^n}$.

• note that $r(A)$ is not a norm in general since $r\left(\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\right) = 0$.

• Proposition. $\sigma(A) \neq \emptyset$.

proof: $R_\lambda = (\lambda I - A)^{-1}$ is analytic in λ on $\rho(A)$. so $\langle x, R_\lambda y \rangle$ is analytic.

Moreover $\langle x, (\lambda I - A)^{-1} y \rangle = \langle x, \frac{1}{\lambda} (I - \frac{A}{\lambda})^{-1} y \rangle \rightarrow 0$ as $|\lambda| \rightarrow \infty$.

If $\rho(A) = \mathbb{C}$, then $\lambda \mapsto \langle x, R_\lambda y \rangle$ is a bounded entire function.

The Liouville theorem implies that $\langle x, R_\lambda y \rangle$ is constant, hence vanishes.

But this implies $R_\lambda \equiv 0$, which is impossible as an inverse.

So $\rho(A) \neq \mathbb{C}$. There must be singularities/poles of R_λ .

This is the spectrum of A .

Spectral theorem for compact self-adjoint operators.

$$A = A^* \in K(H).$$

Let $A \in B(H)$. $R_\lambda = (\lambda I - A)^{-1}$ defined on $\rho(A) \subset \mathbb{C}$ open set.

$\sigma(A) = \mathbb{C} \setminus \rho(A)$ is closed set, spectrum of A and $\sigma(A) \neq \emptyset$.

$\sigma(A) = \sigma_p(A) \cup \sigma_c(A) \cup \sigma_r(A)$ by definition.

Lemma. Assume $A = A^*$. Then $\lambda \in \sigma_p(A) \Rightarrow \lambda = \bar{\lambda}$ (i.e., $\lambda \in \mathbb{R}$).

Moreover eigenvectors for different eigenvalues are orthogonal.

proof: $\lambda \langle x, x \rangle = \langle x, \lambda x \rangle = \langle x, Ax \rangle = \langle Ax, x \rangle = \bar{\lambda} \langle x, x \rangle$.

when $\|x\| \neq 0$, we deduce $\lambda = \bar{\lambda}$.

Assume $Ax = \lambda x$ and $Ay = \mu y$ for $\lambda \neq \mu$. Then:

$$\lambda \langle x, y \rangle = \langle \lambda x, y \rangle = \langle Ax, y \rangle = \langle x, Ay \rangle = \mu \langle x, y \rangle.$$

so, when $\lambda \neq \mu$, we have $\langle x, y \rangle = 0$ $\square$

Definition. $\mathcal{M} \subset \mathcal{H}$ is invariant subspace of A if $A\mathcal{M} \subset \mathcal{M}$. Divide and Conquer tool.

Lemma $A = A^*$ and $A\mathcal{M} \subset \mathcal{M} \Rightarrow A\mathcal{M}^\perp \subset \mathcal{M}^\perp$

proof: $x \in \mathcal{M}^\perp$ and $y \in \mathcal{M}$. $\langle y, Ax \rangle = \langle \underbrace{Ay}_{\in \mathcal{M}}, \underbrace{x}_{\in \mathcal{M}^\perp} \rangle = 0 \quad \forall y \in \mathcal{M}$. So $Ax \in \mathcal{M}^\perp$.

Proposition. $\lambda \in \sigma_r(A) \Rightarrow \bar{\lambda} \in \sigma_p(A^*)$. Note $(A - \lambda I)^* = A^* - \bar{\lambda} I$.

A bounded.

proof: $\lambda \in \sigma_r \Rightarrow \text{ran}(A - \lambda I)$ not dense. $\mathcal{M} = \overline{\text{ran}(A - \lambda I)} \subsetneq \mathcal{H} = \mathcal{M} \oplus \mathcal{M}^\perp$.

So, $\exists x \in \mathcal{M}^\perp, x \neq 0$. $x \in (\text{ran}(A - \lambda I))^\perp \Rightarrow x \in \text{Ker}(A^* - \bar{\lambda} I)$

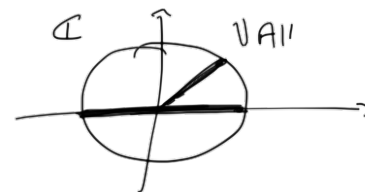
Since $\mathcal{H} = \overline{\text{ran}(A - \lambda I)} \oplus \text{Ker}(A - \lambda I)^*$.

Thus, $A^*x = \bar{\lambda}x$ and $\bar{\lambda} \in \sigma_p(A^*)$.

Corollary. When $A = A^*$, then $\sigma_r(A) = \emptyset$.

Indeed, $\lambda \in \sigma_r(A) \Rightarrow \bar{\lambda} \in \sigma_p(A^*) = \sigma_p(A)$! impossible.
 $\Rightarrow \lambda$ real-valued !

Lemma. Assume $A = A^*$. Then $\sigma(A) \subset [-\|A\|, \|A\|] \subset \mathbb{R}$.



proof. Assume $\lambda = a + ib \in \sigma(A)$ with $b \neq 0$.

$\lambda \notin \sigma_p(A)$ since $\lambda \in \sigma_p(A) \Rightarrow \lambda = \bar{\lambda}$. $\lambda \notin \sigma_r(A)$ since $\sigma_r(A) = \emptyset$.

So, $\lambda \in \sigma_c(A)$ ie $(\lambda I - A)$ one-to-one not onto with dense range.

Look at $\|(A - \lambda)x\|^2 = \|(A - a)x + ibx\|^2 = \|(A - a)x\|^2 + \|bx\|^2$ since $A = A^*$.
(check!)

Thus $\|(A - \lambda)x\|^2 \geq b^2 \|x\|^2$ with $b^2 > 0$.

From proposition on a priori estimates, we deduce $\text{ran}(A - \lambda)$ closed
and $\text{Ker}(A - \lambda) = \{0\}$. But $(\lambda I - A)$ with dense closed range is onto.

This is a contradiction. So, $b = 0$ and $\lambda \in \mathbb{R}$. We know $|\lambda| \leq \|A\|$.
□

Proposition. Let A be compact and self-adjoint.

Then a non-zero eigenvalue of A has finite multiplicity.

And a countably infinite set of eigenvalues of A that do not vanish admits 0 as only accumulation point.

Proof. $\left| \begin{array}{l} A u_k = \lambda u_k \quad k \in \mathbb{N} \quad \lambda \neq 0 \\ u_k \text{ basis of eigenspace, } \|u_k\| = 1 \end{array} \right|$ Then $A u_{\varphi(k)} \rightarrow y$ since A compact.
 $y \leftarrow A u_{\varphi(k)} = \lambda u_{\varphi(k)}$ not Cauchy. Contradiction.

Now assume $\{\lambda_n\}$ bounded with $0 \neq \lambda_{\varphi(n)} \rightarrow \lambda$. Assume $\lambda \neq 0$.

$A e_{\varphi(n)} = \lambda_{\varphi(n)} e_{\varphi(n)}$ for some $e_{\varphi(n)}$. Define $f_{\varphi(n)} = \frac{e_{\varphi(n)}}{\lambda_{\varphi(n)}} \leftarrow \neq 0$

then $A f_{\varphi(n)} = e_{\varphi(n)}$. Assume $\left| \frac{1}{\lambda_{\varphi(n)}} \right| \leq M$. Then $f_{\varphi(n)}$ bounded

A compact implies $A f_{\varphi(n)} \rightarrow y$. But $e_{\varphi(n)}$ not Cauchy.

So, there is no such M , i.e., $\left| \frac{1}{\lambda_{\varphi(n)}} \right| \rightarrow \infty$ and hence $\lambda = 0$ $\square$

. Spectral Theorem for compact self-adjoint operators.

. $A: \mathcal{H} \rightarrow \mathcal{H}$ compact with $A = A^*$.

. Then there is an o.n.b. of $\mathcal{H}$ composed of eigenvectors of A .

. The non-zero eigenvalues of A form a (at most) countably infinite set $\{\lambda_k\}_{k \in \mathbb{N}}$ of real numbers such that $A = \sum_{k \in \mathbb{N}} \lambda_k P_k$ with P_k orthogonal projectors.

. Notation: $A u_k = \lambda_k u_k$, $\|u_k\|=1$ write $P_k = u_k \otimes u_k$ rank-one projector
 $P_k x = \langle u_k, x \rangle u_k$.

. If $\lambda_k = \lambda_{k+1} = \dots = \lambda_{k+m} = \lambda \neq 0$, m finite, then we define

$$P_\lambda = \sum_{l=1}^m P_{k+l} = \sum_{l=1}^m u_{k+l} \otimes u_{k+l}.$$

$$\rightarrow A = \sum_{\lambda \in \sigma_p(A)} \lambda P_\lambda.$$

Proof of theorem. (i) Either $\|A\|$ or $-\|A\|$ belongs to $\sigma_p(A)$. 

Assume $\|A\| \neq 0$ (otherwise trivial). $\|A\| = \sup_{\|x\|=1} |\langle x, Ax \rangle| = \lim_{\substack{n \rightarrow \infty \\ \|x_n\|=1}} |\langle x_n, Ax_n \rangle|$.

x_n : maximizing sequence. x_n is bounded and up to a subsequence, $\langle x_n, Ax_n \rangle \rightarrow \lambda$ with $|\lambda| = \|A\| > 0$.

Since x_n is bounded and A is compact, $Ax_{\varphi(n)} \rightarrow y \in \mathcal{H}$. $y \neq 0$.

It remains to verify that $Ay = \lambda y$.

$$\|(A - \lambda)y\|^2 = \lim_{n \rightarrow \infty} \|(A - \lambda)Ax_{\varphi(n)}\|^2 \leq \|A\|^2 \lim_{n \rightarrow \infty} \|(A - \lambda)x_{\varphi(n)}\|^2$$

$$\text{A.S.A.} \Rightarrow \leq \|A\|^2 \lim_{n \rightarrow \infty} \left(\underbrace{\|Ax_{\varphi(n)}\|^2}_{\leq \|A\|^2} + \underbrace{\lambda^2 \|x_{\varphi(n)}\|^2}_{=1} - 2\lambda \underbrace{\langle x_{\varphi(n)}, Ax_{\varphi(n)} \rangle}_{\rightarrow \lambda} \right)$$

$$\leq \|A\|^2 (\|A\|^2 + \lambda^2 - 2\lambda^2) = 0$$

So, $\lambda \in \sigma_p(A)$ for $\lambda = \|A\|$ or $\lambda = -\|A\|$.

(2). Define $\mathcal{N}_1 = \mathcal{H}$ and $A_1 = A$.

• $A_1 e_1 = \lambda_1 e_1 \in \mathcal{N}_1$ $\lambda_1 = \pm \|A\|$, $e_1 = y$ from step 1.

• $\mathcal{M}_2 = \mathbb{R} e_1$ and $\mathcal{N}_1 = \mathcal{M}_2 \oplus \mathcal{N}_2$ with $\mathcal{N}_2 = \mathcal{M}_2^\perp$

$$A_1 = \left(\begin{array}{c|c} \lambda_1 & 0 \\ \hline 0 & A_2 \end{array} \right) \quad \begin{array}{c} \underbrace{\hspace{1cm}}_{\mathcal{M}_2} \quad \underbrace{\hspace{1cm}}_{\mathcal{M}_2^\perp} \end{array}$$

• $\mathcal{M}_2$ invariant of A s.A. $\Rightarrow \mathcal{M}_2^\perp = \mathcal{N}_2$ as well: $A_1 \mathcal{M}_2 \subset \mathcal{M}_2$
 $A_1 \mathcal{M}_2^\perp \subset \mathcal{M}_2^\perp$

So, we define $A_2 = A_1|_{\mathcal{N}_2}$ with $\|A_2\| \leq \|A_1\|$.

• we can repeat step 1: $A_2 e_2 = \lambda_2 e_2$ $|\lambda_2| = \|A_2\| \leq |\lambda_1|$, $e_1 \perp e_2$.

• By induction, we construct $\left| \begin{array}{l} \mathcal{M}_n = \text{span} \{ e_1 \dots e_n \} \quad \mathcal{N}_n = \mathcal{M}_n^\perp \\ A_n = A|_{\mathcal{N}_n} = A_{n-1}|_{\mathcal{N}_n} \end{array} \right. \quad n \geq 1.$

(3). If $A_{n+1} = 0$, then $A = \sum_{k=1}^n \lambda_k e_k \otimes e_k =: \sum_{k=1}^n \lambda_k P_k$.

• $A|_{\mathcal{N}_n} = 0$ with $\mathcal{N}_n = \mathcal{M}_n^\perp$ kernel of A a closed subspace of $\mathcal{H}$.

• $\mathcal{N}_n$ admits a basis of eigenvectors $\{f_k\}$. $(\{e_1, \dots, e_n\}, \{f_k\})$ o.n.b. of $\mathcal{H}$.

(4) If $A_n \neq 0 \forall n$ (only in infinite dimension) then $\{\lambda_n\}$ infinite sequence of e.v.'s of A with $0 < |\lambda_{n+1}| \leq |\lambda_n|$.

We know 0 only accumulation point of e.v. for compact operator.

Thus $\lambda_n \rightarrow 0$. $A = \sum_{k=1}^n \lambda_k e_k \otimes e_k + A_{n+1}$ $\|A_{n+1}\| = |\lambda_{n+1}| \rightarrow 0$

So $\lim_{n \rightarrow \infty} \|A - \sum_{k=1}^n \lambda_k e_k \otimes e_k\| = 0$ and, hence $A = \sum_{k=1}^{\infty} \lambda_k P_k$.

(5) It remains to understand the range of A , which is not closed, and its kernel. clearly, $\text{ran } A \subset [\{e_k\}]$. Let $x \in [\{e_k\}]$ i.e. $x = \sum_{k=1}^{\infty} b_k e_k$ with $\sum_{k=1}^{\infty} |b_k|^2 = \|x\|^2$

Then $Ax = \sum_k \lambda_k P_k x = \sum_k \lambda_k b_k e_k = \sum_k c_k e_k$ with $\sum_{k=1}^{\infty} \frac{|c_k|^2}{\lambda_k^2} = \|x\|^2 < \infty$.

So, $\text{Ran } A = \left\{ \sum_{k=1}^{\infty} c_k e_k, \sum_k \left| \frac{c_k}{\lambda_k} \right|^2 < \infty \right\}$ Picard criterion.

Note $\text{Ran } A \subsetneq \overline{\text{Ran } A} = [\{e_k\}_{k \geq 1}] =: \mathcal{M}$.

Now $\mathcal{H} = \mathcal{M} \oplus \mathcal{M}^{\perp}$. $x \in \mathcal{M}^{\perp} \Rightarrow Ax = A_n x \forall n \Rightarrow Ax = 0$

$\left| \begin{array}{l} \mathcal{M}^{\perp} = \text{Ker } A. \\ \{f_j\} \text{ o.n.b. of } \mathcal{M}^{\perp} \end{array} \right| \left| \begin{array}{l} A = \sum_{j=1}^{\infty} \lambda_j P_j \quad P_j = e_j \otimes e_j \text{ with} \\ (\{e_k\}, \{f_j\}) \text{ o.n.b. of } \mathcal{H} \end{array} \right| \quad \square$